MEDI SEARCH

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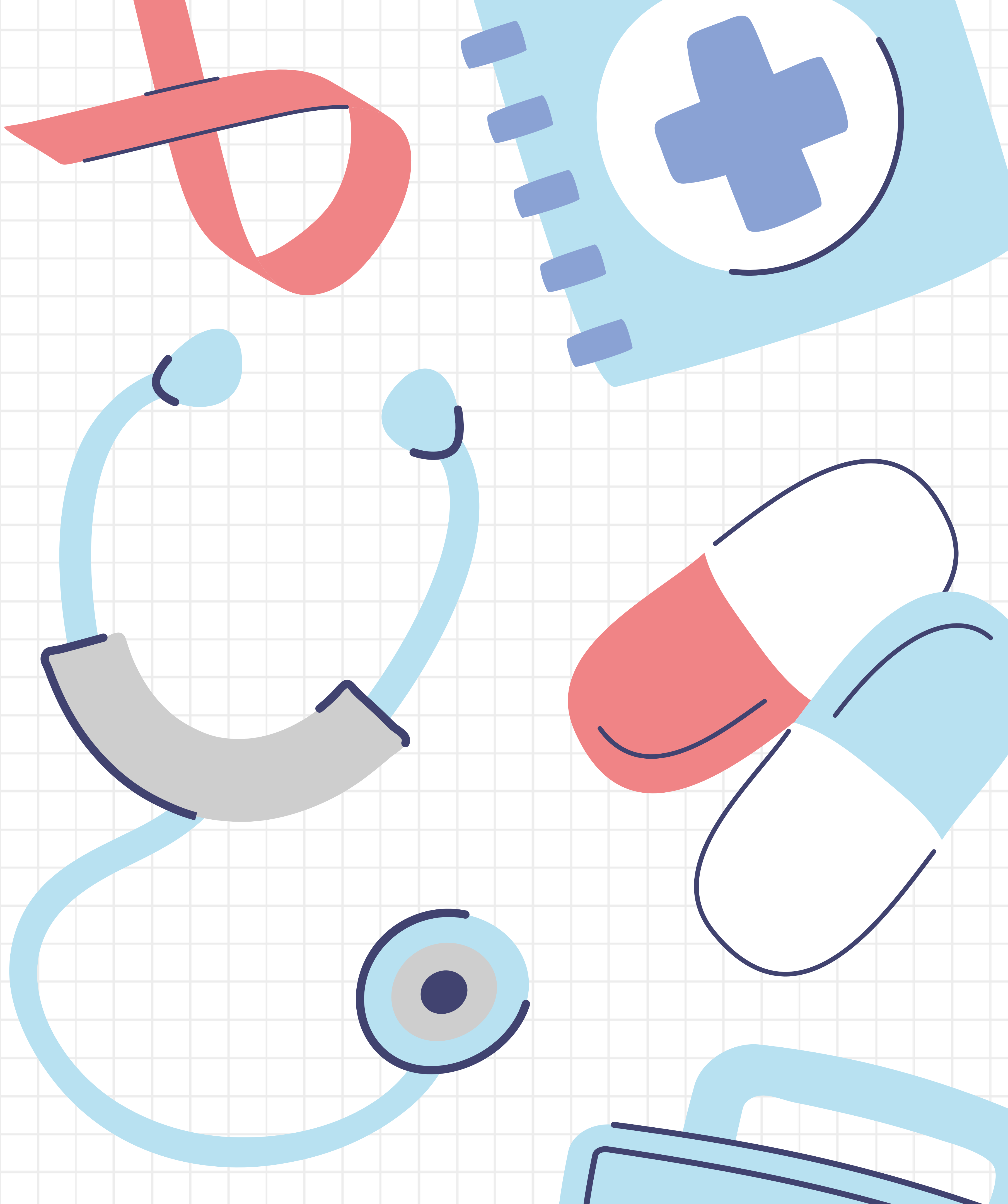


Abstract

MediSearch: Intelligent Patient Retrieval System Using Advanced String Algorithms is a Data Structures and Algorithms (DSA)-based project developed to enable fast and efficient searching of patient records in large healthcare databases. Unlike traditional hospital management systems, this project focuses on intelligent information retrieval using advanced search algorithms. The system allows users to search patient records by patient ID, name, disease, symptoms, diagnosis, or medicines. It utilizes HashMap for quick patient ID lookup, Trie for autocomplete and prefix-based search, Knuth-Morris-Pratt (KMP) algorithm for efficient pattern matching, and Edit Distance (Dynamic Programming) to detect and correct misspelled medical terms. These algorithms reduce search time, improve search accuracy, and provide relevant results even for incomplete or incorrect queries. MediSearch demonstrates the practical application of DSA concepts in solving real-world healthcare challenges by delivering a scalable, accurate, and intelligent patient retrieval system that enhances decision-making and improves the overall efficiency of medical record searching.

Problem Statement

- Hospitals store large amounts of patient data such as medical history, diagnoses, prescriptions, laboratory reports, and appointments.
- Doctors may find it difficult and time-consuming to retrieve the required patient information quickly.
- Traditional record-search methods may not efficiently identify relevant or similar patient cases.
- A searchable patient repository is needed to provide fast, accurate, and intelligent patient information retrieval.
- The system can also help doctors identify similar medical cases to support better clinical decision-making.

Why This Project

Existing System:

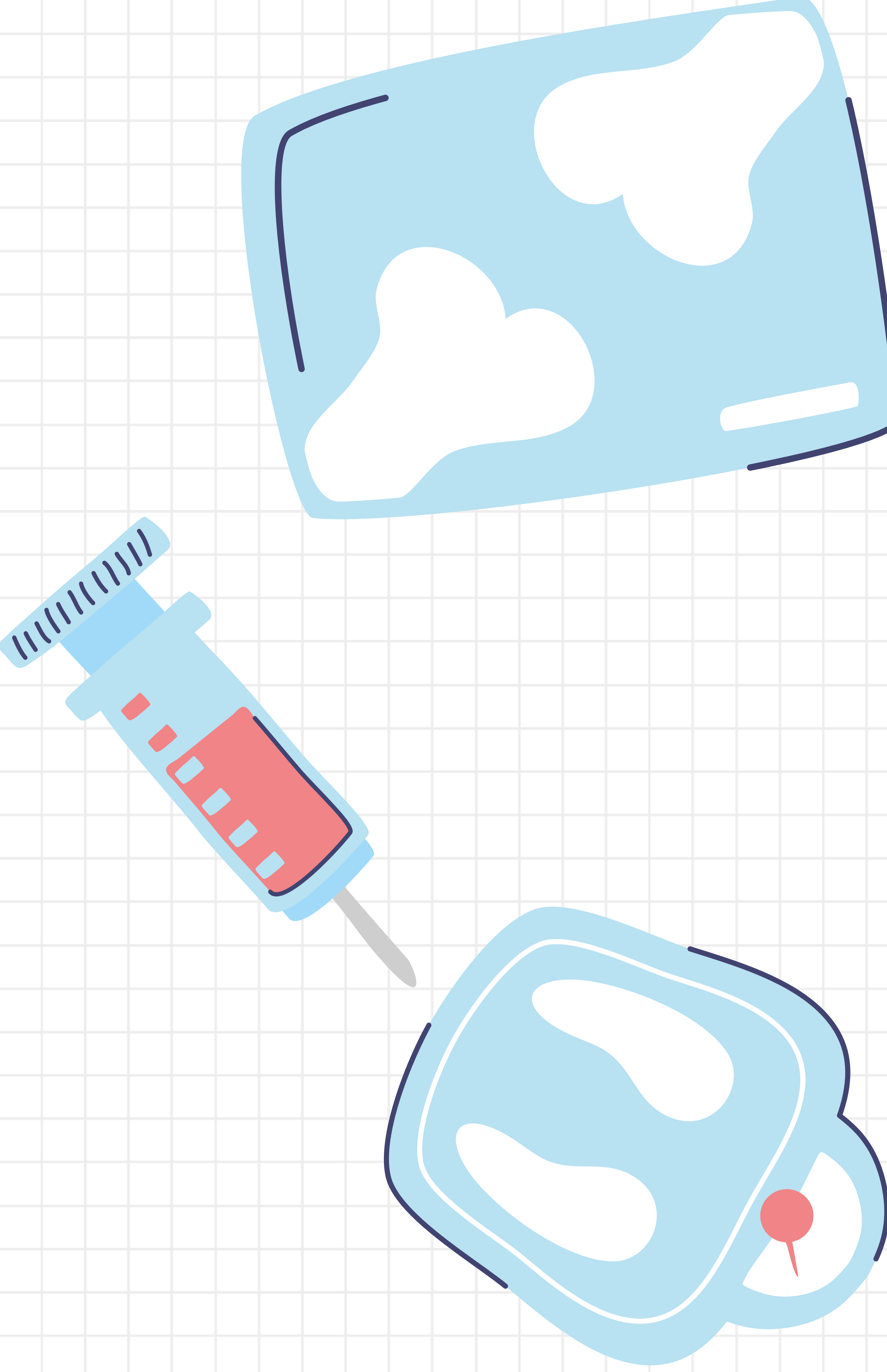
- Patient information scattered across different records
- Slow and difficult patient record searching
- Difficult to find relevant medical history
- No centralized patient information repository
- Time-consuming retrieval of patient records

Proposed System:

- Centralized patient information repository
- Fast and efficient patient record searching
- Organized patient medical records
- Easy access to diagnoses, prescriptions, lab reports, and appointments
- Intelligent retrieval of relevant patient information
- Identification of similar medical cases

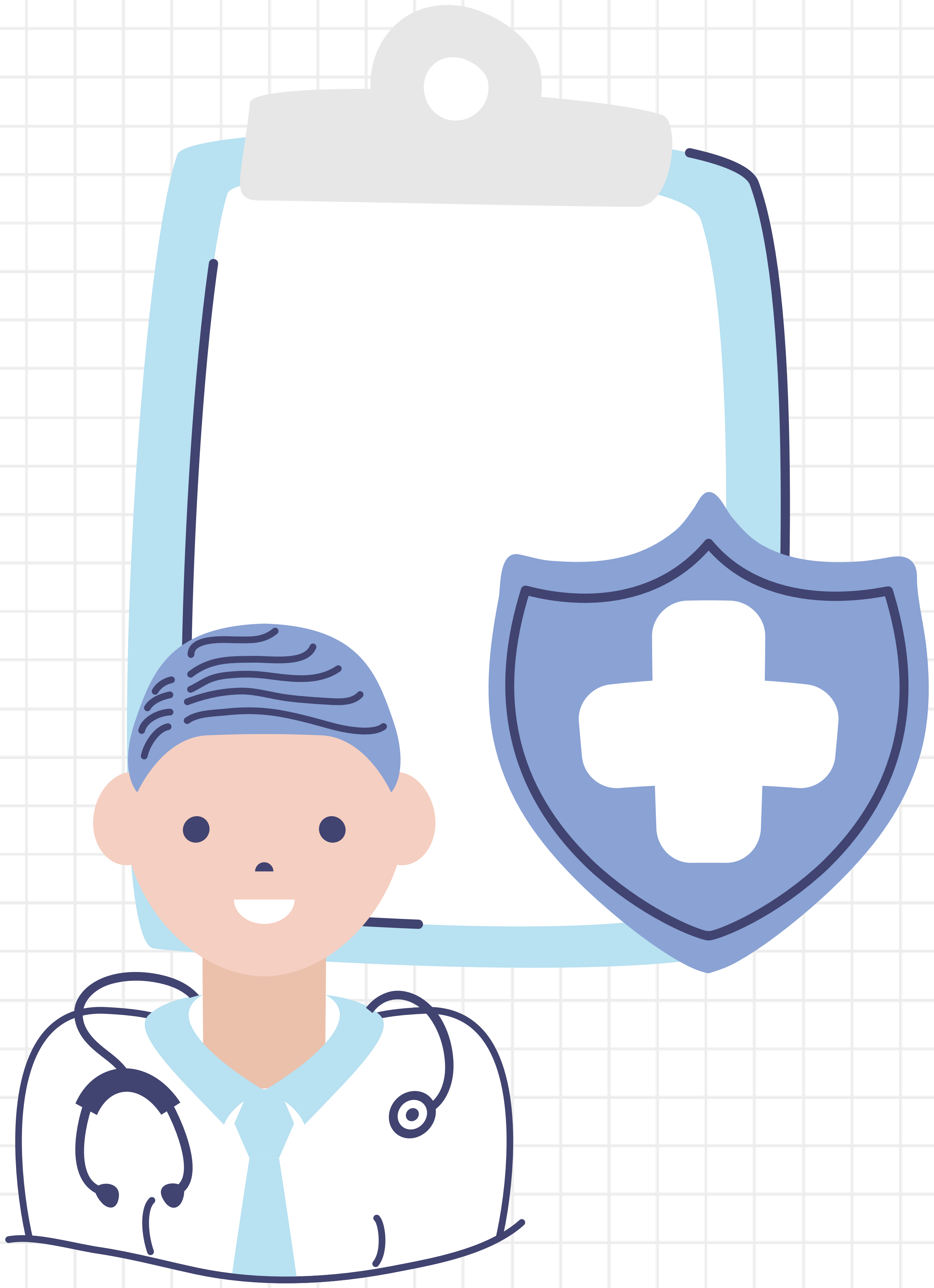
Proposed Functionalities

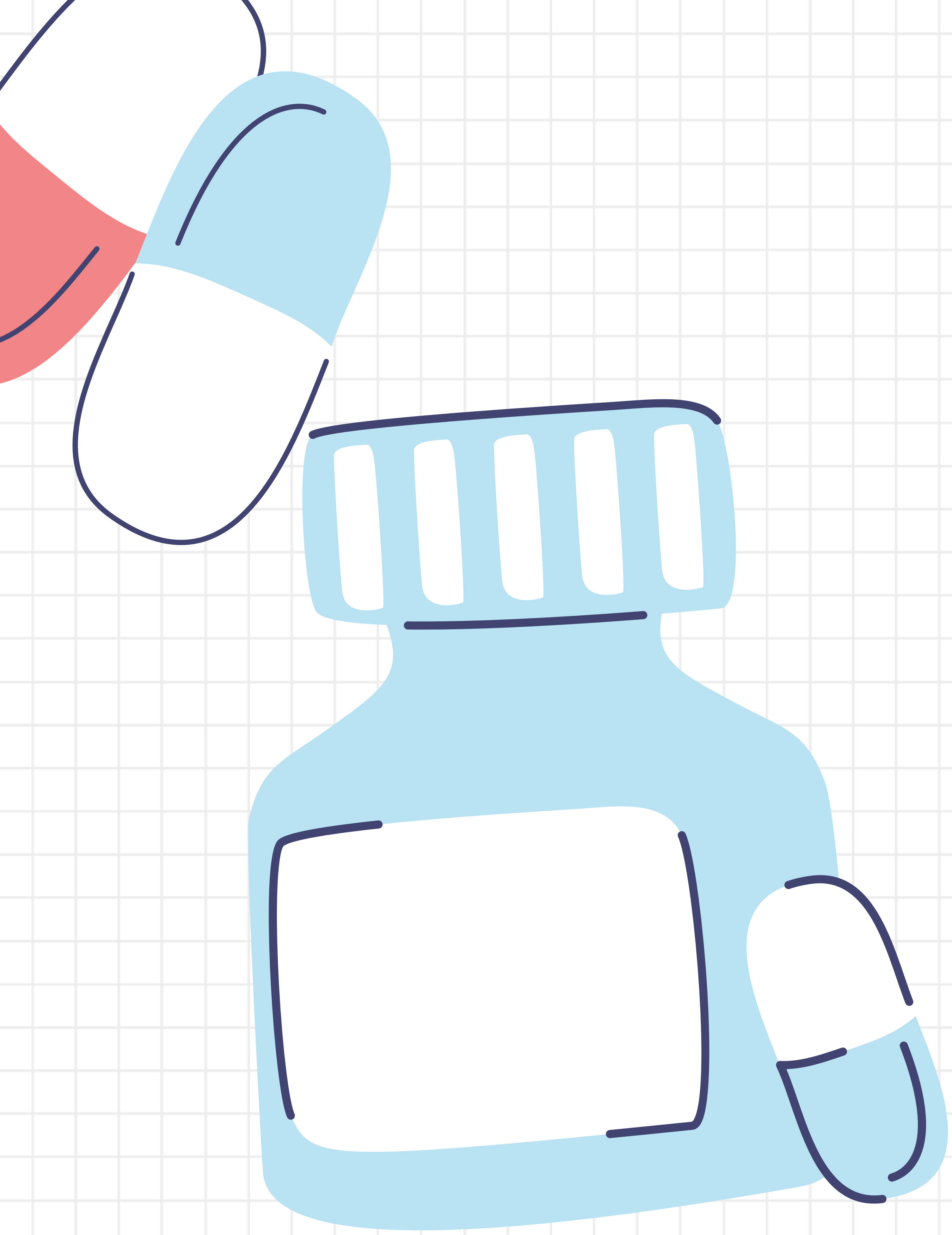
- Functionality Description
- Patient Search Search for patient records using patient ID, name, or other details.
- Diagnosis Search Find patient diagnoses and related medical information.
- Prescription Information View prescribed medicines and treatment details.
- Laboratory Reports Access and retrieve patient laboratory test reports.
- Appointment Information View patient appointment dates and related details.
- Medical History Search Retrieve a patient's previous medical history quickly.



Software Technologies

- Programming Language: Java
- Frontend: Java Swing
- Backend: JDBC
- Database: MySQL
- IDE: IntelliJ IDEA / Eclipse
- Version Control: Git & GitHub



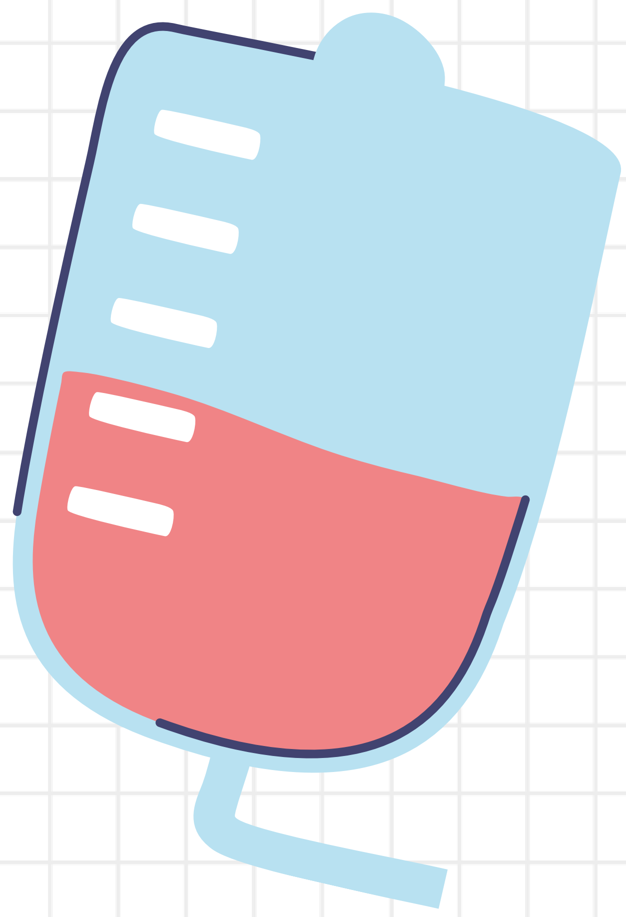


DSA CONCEPTS

Data Structures	Hash Table, Trie, Binary Search Tree
Algorithms	KMP, Rabin-Karp, Boyer-Moore

Expected Outcome

* Fast patient record retrieval
* Accurate patient information search
* Efficient string-based searching
* Reduced search time
* Improved search accuracy



Future Enhancements

* AI-based intelligent patient search
* Natural language search support
* Voice-based patient record search
* Fuzzy matching for misspelled patient information

Thank you!